Breaking the Script Barrier: Automatic De-Romanization of Maghribi Arabizi to Arabic Script in Social Media

Abstract

Social media users in the Maghreb region frequently write dialectal Arabic in Arabizi, a non-standardized Romanized script that mixes Latin characters and numerals. This paper studies automatic de-romanization of Maghribi Arabizi into Arabic script using a controlled comparison of three systems: a deterministic rule-based mapper, a statistical maximum-likelihood word mapper, and a character-level word sequence-to-sequence model. Experiments are conducted on held-out train, validation, and blind-test splits derived from a parallel Maghribi Arabizi-Arabic corpus. We further introduce native-speaker validation of a 300-row blind-test subset to verify the reliability of the gold references, and a separate human review of sampled neural errors. Results show that the statistical MLE model achieves the strongest corpus-level performance, while the neural model improves over the rule-based baseline but remains sensitive to lexical substitution, phonetic ambiguity, and source noise. The study highlights both the promise and the difficulty of de-romanizing low-resource dialectal Arabic under high orthographic variation.

1 Introduction

The expansion of social media platforms has accelerated the use of dialectal Arabic online. In the Maghreb, users frequently write Moroccan, Algerian, and Tunisian varieties in Arabizi, an informal Romanized script that uses Latin letters and numerals to approximate Arabic phonology, such as 7 for ح /ḥ/ and 3 for ع /ʿ/. This practice supports fast informal communication, but it creates a barrier for Arabic NLP systems that expect native Arabic script (Alayba, 2025).

The task studied here is de-romanization: converting Maghribi Arabizi into Arabic script while preserving the dialectal lexical and morphological form rather than translating the sentence into Modern Standard Arabic. This distinction is related to prior Arabic romanization and script-transfer work (Eryani and Habash, 2021), and it matters because a valid Darija form may be penalized by strict string metrics even when it is linguistically acceptable.

Maghribi Arabizi is challenging for three reasons. First, spelling is highly variable: the same token may appear with different vowels, doubled letters, or omitted segments. Second, clitics and function words often fuse into single surface forms. Third, Maghribi social media includes frequent French and Spanish code-switching. We therefore compare three methods of increasing modeling complexity: a hand-built rule system, a word-level MLE mapper, and a neural encoder-decoder model.

2 Dataset Curation and Preprocessing Pipeline

2.1 Corpus Source and Experimental Splits

The parallel data are drawn from the UBC-NLP NileChat Arabizi-Morocco resource distributed on Hugging Face as UBC-NLP/nilechat-arabizi-mor. The dataset card describes the resource as Moroccan Arabic dialect data expressed in Arabizi and requests citation of the NileChat paper (El Mekki et al., 2025). The dataset is released under the Creative Commons Attribution-NonCommercial 4.0 license. For the experiments reported in the accompanying notebook, the final exported split contains 6,436 training sentence pairs, 804 validation pairs, and 805 blind-test pairs. The blind test set is kept isolated until final system comparison.

2.2 Cleaning and Alignment Constraints

The preprocessing pipeline removes empty rows, normalizes whitespace, and relies on aligned source-target text fields. For model-specific training, the neural system extracts word-level pairs only when the Arabizi and Arabic sentences have equal word counts. This constraint produced 79,909 training word pairs and 12,165 validation word pairs, while 5,744 training rows and 713 validation

rows were skipped only for neural word-pair extraction. These rows were not deleted from the corpus or from the blind evaluation.

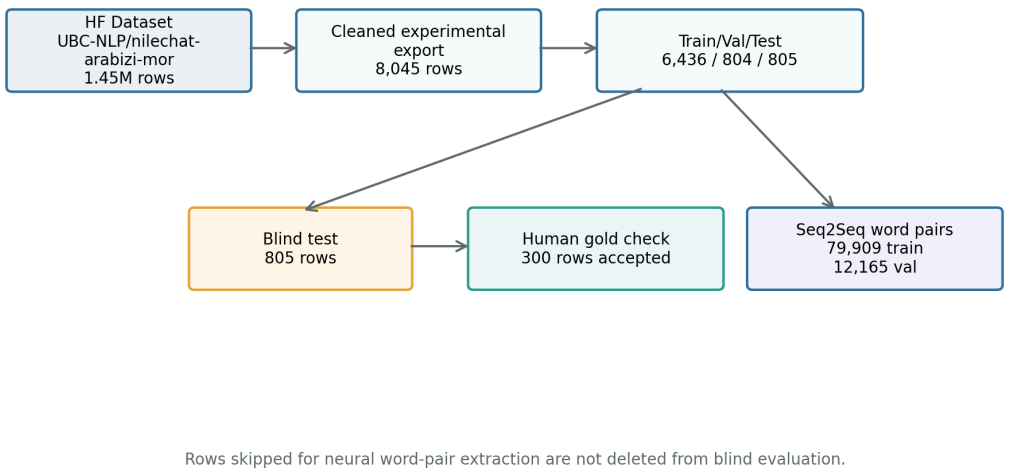


Figure 1: Experimental data flow and evaluation subsets.
The flow shows the source corpus, final train/validation/test splits, the 300-row human gold validation sample, and the word-pair extraction stage used only for the neural model.

2.3 Lightweight Custom Normalization

The experiments do not apply CAMeL Tools script standardization. Instead, the notebook uses lightweight custom normalization: whitespace cleanup, lowercasing for Arabizi inputs, and token-level Arabic normalization during aligned qualitative analysis. This preserves the reference style of the original Arabic targets while reducing minor comparison artifacts. CAMeL Tools remains an important Arabic NLP toolkit (Obeid et al., 2020), but it was not part of the executed preprocessing pipeline.

2.4 Human Gold-Standard Test Curation

To verify the reliability of the blind test references, two native Moroccan Arabic speakers independently annotated a random sample of 300 rows. Their task was to determine whether each Arabizi input and Arabic-script target formed a valid de-romanization pair suitable for evaluation. The annotation therefore served as gold-set quality control rather than as model training data.

Both annotators accepted all 300 rows, yielding 100% observed agreement and 300 curated agreed-accurate rows. Because all annotated instances received the same final validity label, Cohen's kappa is reported with caution: the absence of label variation limits its interpretability as a chance-corrected statistic.

3 System Architecture and Computational Models

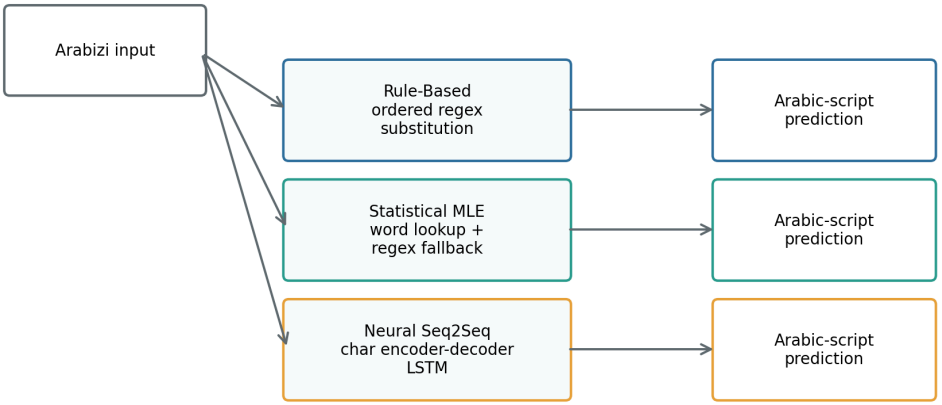


Figure 2: Comparison of the three de-romanization architectures.
The diagram contrasts the rule-based character mapper, the MLE word lookup with regex fallback, and the character-level word Seq2Seq encoder-decoder model.

3.1 Algorithm 1: Deterministic Rule-Based Mapper

The rule-based baseline uses ordered regular-expression substitutions to map common Arabizi symbols and digraphs into Arabic script. Multi-character patterns such as kh, gh, ch, ou, and ee are applied before single-character replacements. Numeral mappings include $7\ to$ ح, $3\ to$ ع, $9\ to$ ق, $5\ to$ خ, $2\ to$ ء, $and\ 6\ to$ ط when present in the input. The system is transparent and deterministic, but it cannot resolve lexical ambiguity or context-dependent spelling.

Rule Category	Arabizi Input	Arabic Script Output
Digit-to-phoneme mapping	7, 3, 9, 2	ح, ع, ق, ء
Complex phoneme mapping	kh, ch/sh, gh	خ, ش, غ
Vowel mapping	a, i, ou/u	ا, ي, و
Fallback consonants	b, t, k, m, n	ب, ت, ك, م, ن

3.2 Algorithm 2: Statistical Maximum Likelihood Estimation Mapper

The statistical baseline estimates word-level mappings from the training split. For each Arabizi token, the model counts aligned Arabic target tokens and selects the most frequent target according to maximum likelihood. If a token is unseen at test time, the model falls back to the rule-based mapper. This hybrid design gives the system access to observed lexical regularities while retaining coverage for out-of-vocabulary tokens, a common challenge in dialectal Arabic processing (Aminian et al., 2014).

Formally, for an Arabizi token Waz and candidate Arabic token War, the model estimates $P(w_{ar} \mid w_{az})$ from training counts and selects the candidate with maximum conditional probability. In the executed notebook, the MLE table contains 46,901 unique Arabizi tokens.

$$P(w_{ar} \mid w_{az}) = C(w_{az}, w_{ar}) \,/\, \Sigma_{w' \in Var}\, C(w_{az}, w')$$

3.3 Algorithm 3: Character-Level Word Seq2Seq Model

The neural model is a TensorFlow/Keras encoder-decoder LSTM trained at the word level. Each Arabizi word is encoded as a sequence of one-hot characters, and the decoder generates Arabic characters using start and end tokens. The final notebook uses 79 encoder characters, 142 decoder characters, a maximum encoder word length of 23, a maximum decoder word length of 22, a latent dimension of 128, batch size 128, and up to 50 training epochs with validation monitoring and checkpointing.

This model tests whether a neural sequence model can learn character-level transliteration patterns from the available aligned word pairs. Unlike a pretrained Transformer, it does not use external Arabic pretraining or subword attention. This makes it a useful controlled neural baseline, but also limits its ability to model longer sentence context and rare lexical substitutions.

The decoder is optimized with categorical cross-entropy over the Arabic character vocabulary:

$$L = -\frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{|V_{ar}|} y_{t,c}\log(\hat{y}_{t,c})$$

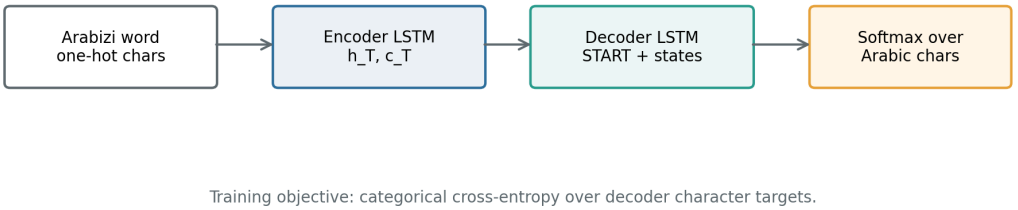


Figure 3: Word-level character Seq2Seq encoder-decoder model.
The diagram shows Arabizi word characters entering an LSTM encoder, decoder state transfer, and Arabic character generation using start/end tokens.

4 Experimental Setup

All systems are trained only on the training split and evaluated on the untouched blind test set. The rule-based model does not learn parameters. The MLE model is trained from aligned word counts in the training data. The neural Seq2Seq model is trained on extracted aligned word pairs, with validation loss monitored during training.

Evaluation uses SacreBLEU with international tokenization, individual 1-gram through 4-gram precision scores, and position-wise exact word match. The exact-match metric is deliberately named position-wise because it penalizes token shifts even when a human may judge the output as partially acceptable.

5 Quantitative Evaluation and Empirical Results

5.1 Global Benchmark Performance

Table 1 reports blind-test performance for the three systems. The strongest corpus-level model is the statistical MLE mapper, which substantially outperforms both the rule-based baseline and the neural word-level Seq2Seq model. The neural model improves over the rule-based baseline, but it does not surpass the lexical coverage provided by the MLE table.

Table 1: Blind-test transliteration benchmark results.

Model	**SacreBLEU**	**Position-wise Exact Word Match (%)**
Rule-Based	7.24	8.69
Statistical MLE	35.44	16.76

Neural Seq2Seq	9.62	9.42

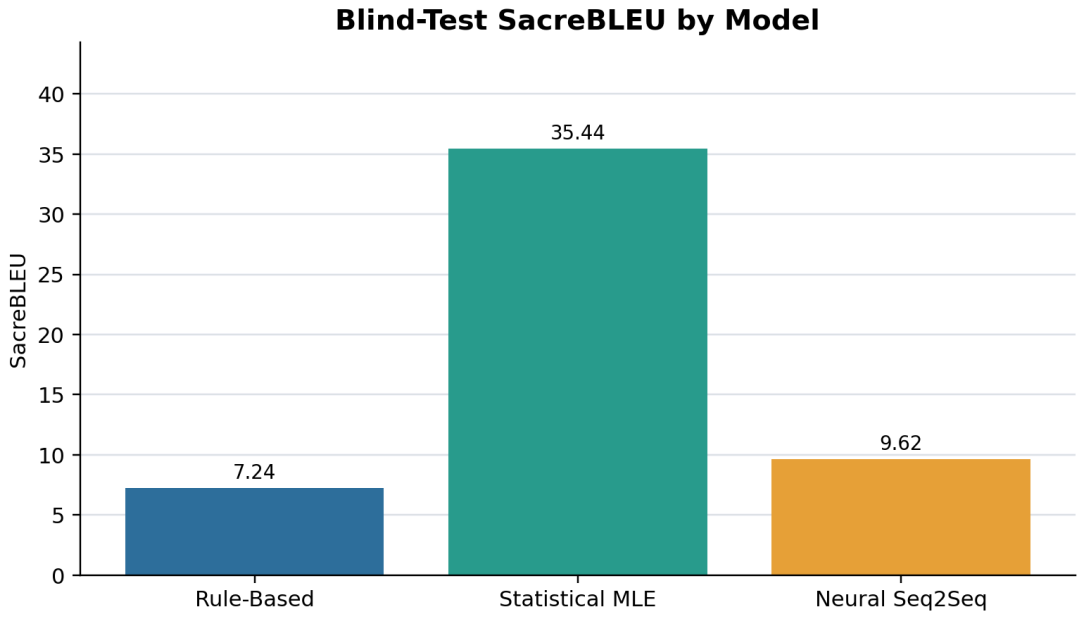


Figure 4: Blind-test SacreBLEU by model.
The bar chart shows the relative benchmark performance of the Rule-Based, Statistical MLE, and Neural Seq2Seq systems.

5.2 N-gram Precision Decay

Table 2 decomposes SacreBLEU into individual n-gram precision scores. The rule-based system loses precision rapidly because character substitution alone cannot recover lexical or syntactic structure. The MLE model has the highest 1-gram precision and remains strongest through 4-grams in this experiment, indicating that many blind-test tokens are covered by observed training mappings. The neural model sits between the two systems for 1-gram precision but drops sharply for longer n-grams, reflecting the difficulty of composing accurate sentence outputs from independently decoded words.

Table 2: Stepwise n-gram overlap precision decay (%).

Model	**1-gram**	**2-gram**	**3-gram**	**4-gram**
Rule-Based	35.62	12.61	4.44	1.58
Statistical MLE	67.80	43.78	29.12	19.49
Neural Seq2Seq	43.93	16.44	6.30	2.43

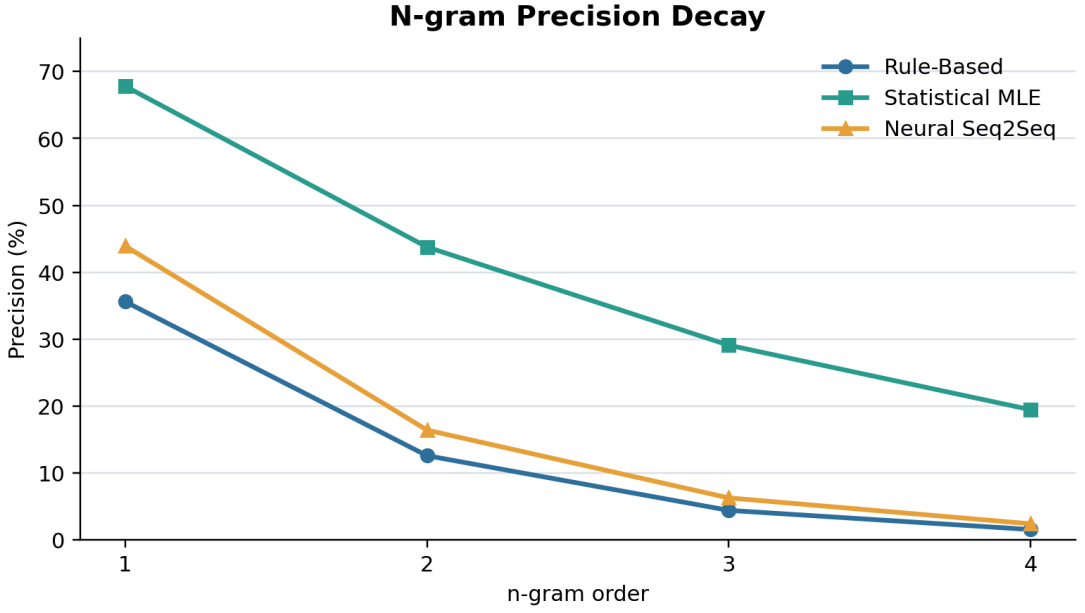


Figure 5: N-gram precision decay across model classes.
The line chart tracks 1-gram through 4-gram precision for all three systems.

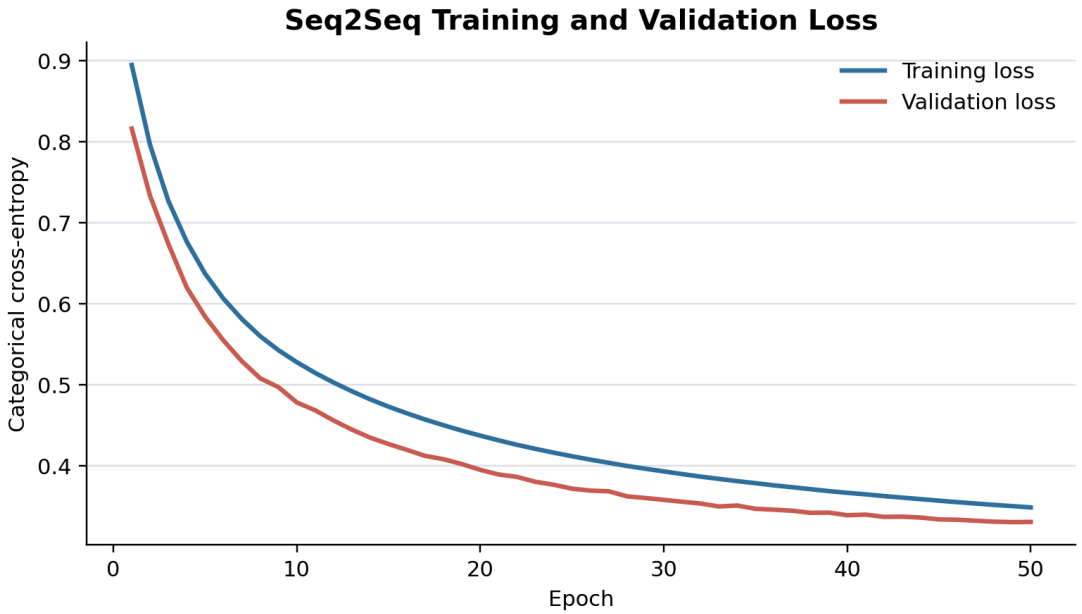


Figure 6: Neural Seq2Seq training and validation loss.
The curve plots categorical cross-entropy across training epochs.

5.3 Human-Curated Subset Evaluation

The same models are also evaluated on the 300-row human-curated gold subset. The relative ranking remains stable: MLE obtains the strongest BLEU, followed by the neural Seq2Seq model and the rule-based system. This suggests that the main benchmark trend is not an artifact of obviously invalid reference pairs.

Table 3: Human-curated gold-subset benchmark results.

Model	Curated SacreBLEU	Curated Position-wise Exact Word Match (%)
Rule-Based	7.34	8.84
Statistical MLE	35.58	16.73
Neural Seq2Seq	9.43	9.32

6 Human-in-the-Loop Qualitative Validation

6.1 Error Review Protocol

A separate human validation stage examines 50 aligned neural error cases sampled from the blind-test outputs. The error extraction uses sequence alignment to reduce false penalties caused by simple word-position shifts. Annotators mark each case as a true model error, acceptable variant/not a true model error, or uncertain, and assign a primary error type.

6.2 Human Judgment Summary

Of the 50 reviewed neural error cases, 42 were judged true model errors, 6 were judged acceptable variants or not true model errors, and 2 remained uncertain. This confirms that automatic string metrics can over-penalize some dialectal or noisy cases, but it does not justify replacing the corpus-level BLEU scores with a manually adjusted BLEU score unless such a metric is explicitly computed.

Table 4: Human validation of sampled neural errors.

Human Judgment	Count
True model error	42
Acceptable variant / not model error	6
Uncertain	2

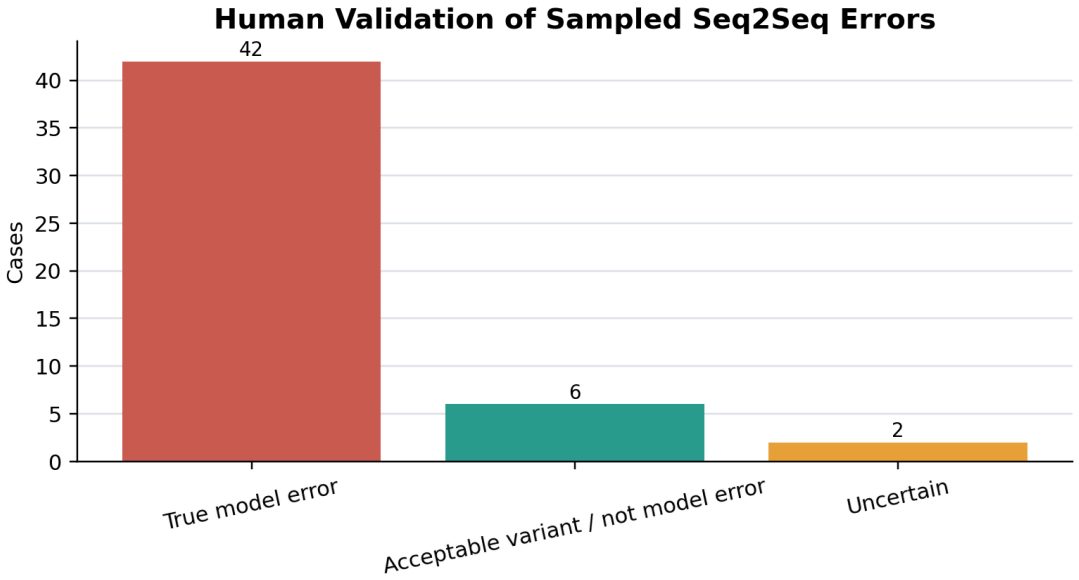


Figure 7: Human judgment distribution for sampled neural errors.
The chart summarizes true model errors, acceptable variants or non-errors, and uncertain cases.

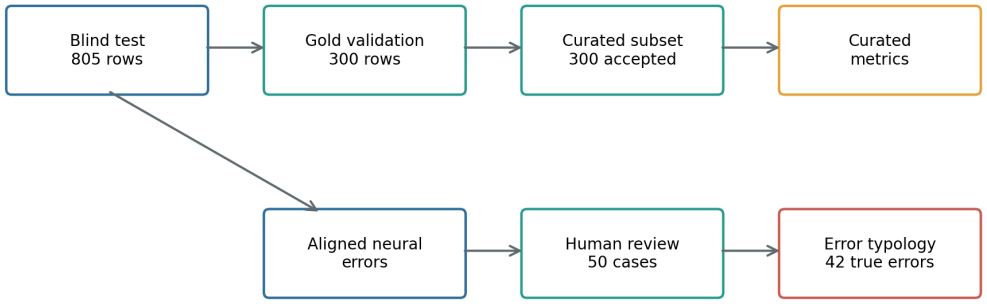


Figure 8: Human-in-the-loop evaluation flow.
The flowchart shows gold-set validation, curated subset evaluation, aligned neural error sampling, and human error review.

6.3 Primary Error Typology

Among the 42 true neural model errors, the dominant primary error type is lexical substitution, followed by phonetic ambiguity. Source noise appears less frequently as a primary label, though it may still contribute as a secondary factor in individual cases.

Table 5: Primary human-validated Seq2Seq error types among true model errors.

Primary Human Error Type	Count
Lexical substitution	22
Phonetic ambiguity	19
Source noise	1

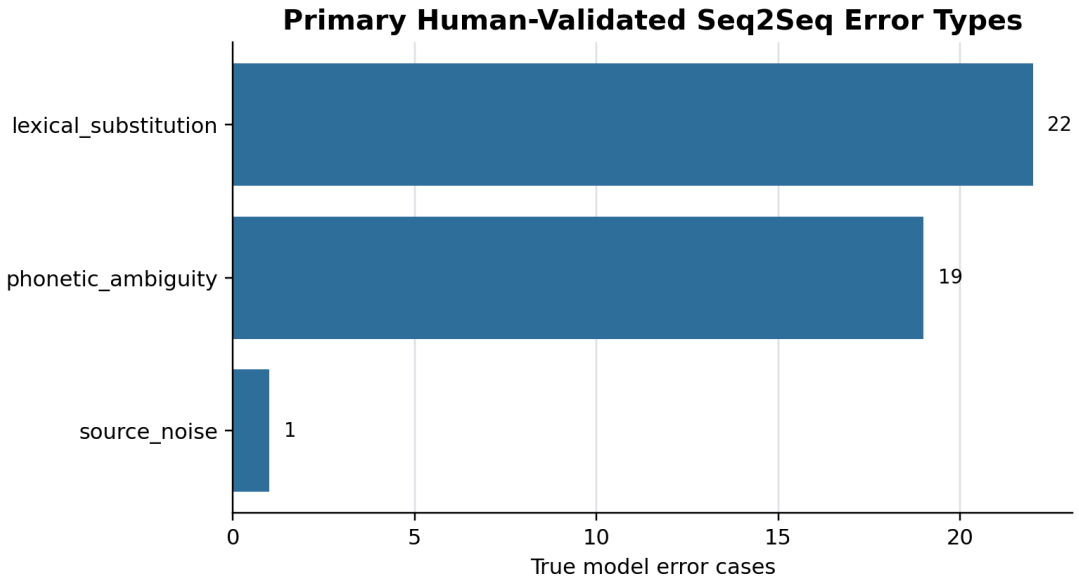


Figure 9: Primary human-validated Seq2Seq error types.
The bar chart shows primary error types among true neural model errors.

7 Qualitative Linguistic Analysis

The aligned qualitative analysis focuses on phenomena that are central to Maghribi Arabizi: numerical transliteration such as 7 -> ح and 3 -> ع, clitic attachment, source-side noise, punctuation, and lexical ambiguity. Some cases that appear incorrect under exact token matching are better interpreted as reference or source-noise issues, which motivates separating automatic metrics from human linguistic interpretation.

- Numerical transliteration: the model often handles frequent mappings such as 3 -> ع, but mixed numeric expressions such as dates and times remain difficult.
- Clitic attachment: fused forms with l-, w-, b-, and f- can create segmentation ambiguity in both prediction and evaluation.
- Lexical substitution: the neural model sometimes produces a fluent but unrelated Arabic word, especially for short high-frequency Arabizi tokens.
- Source noise: non-standard or abbreviated Arabizi input can make a strict gold comparison unreliable without human review.

8 Related Work

Romanization and script conversion have long been studied in Arabic NLP, including work on Arabic bibliographic romanization, dialectal lexicons, and tools for Arabic analysis (Eryani and Habash, 2021; Diab et al., 2014; Obeid et al., 2020). Morphological analyzers such as MADAMIRA illustrate the depth of Arabic-script analysis available for standard and dialectal Arabic once text is represented in native script (Pasha et al., 2014). Prior work also shows that romanization can help or hurt cross-script transfer depending on whether lexical and morphological boundaries are preserved (Amrhein and Sennrich, 2020; Purkayastha et al., 2023). The present work differs by focusing on Maghribi Arabizi de-romanization and by comparing transparent rule-based, statistical, and neural baselines under human-validated evaluation.

9 Conclusion and Future Directions

This paper presented a three-system comparison for Maghribi Arabizi de-romanization. The statistical MLE mapper achieved the strongest blind-test performance, demonstrating the value of high-coverage observed word mappings for this dataset. The neural word-level Seq2Seq model improved over the deterministic rule baseline but did not surpass MLE, indicating that a small character-level LSTM is not sufficient to capture the full lexical and contextual complexity of Maghribi social media text.

This study focuses on the transliteration of Maghribi Arabizi into Arabic script as a foundational step toward improved processing of Maghribi Arabic in NLP. Future work will explore the more challenging task of converting Maghribi dialect written in Arabic script into Modern Standard Arabic, with the aim of enabling stronger normalization pipelines and better integration with downstream Arabic language technologies.

Limitations

The main limitation is that the neural model reported here is a controlled word-level LSTM baseline rather than a pretrained contextual transformer. Its performance should therefore not be interpreted as the ceiling for neural de-romanization. The word-level training procedure also excludes rows with unequal source-target word counts from neural pair extraction, which biases the neural training data toward easier alignments. Automatic metrics such as BLEU and position-wise exact match remain sensitive to spelling variation and word alignment. Finally, the gold-set annotation sample produced no negative labels, so Cohen's kappa is not very informative despite perfect observed agreement.

Ethical Considerations and AI-Use Disclosure

The dataset consists of social-media-style language, which may contain personal, noisy, or sensitive content. The experiments operate on text pairs without attempting to identify users, infer personal attributes, or profile individuals. Future public release of derived resources should respect the licensing and privacy conditions of the original corpus and minimize personally identifying content.

Generative AI tools were used for code debugging, editorial feedback, and manuscript revision support. All experimental claims, reported values, and methodological descriptions were checked against the executed notebook by the author.

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